

Language of logic



1

a $\sim$

b $\wedge$

c $\vee$

d $\implies$

e $\iff$

2

a $\sim p$

b $p \wedge q$

c $\sim q \vee \sim p$

d $\sim q \iff \sim p$

e $\sim p \implies \sim q$

f $\sim q \wedge p$

3

a $\sim q \iff p$

b $p \implies \sim q$

c $\sim p \wedge q$

d $\sim p \wedge \sim q$

e $\sim (p \vee q)$

f $\sim (\sim q \implies p)$

4

a James does not have a smartphone.

b Dave has a laptop and James has a smartphone.

c If Dave does not have a laptop, then James has a smartphone.

d Dave does not have a laptop or James does not have a smartphone.

e It is false that Dave has a laptop and James has a smartphone.

5

a Quentin does not have a rabbit.

b Quentin has a rabbit or Maximilian has a dog.

c Quentin does not have a rabbit if and only if Maximilian does not have a dog.

d It is false that Quentin has a rabbit or Maximilian has a dog.

e Quentin does not have a rabbit and Maximilian does not have a dog.

6

a $(p \wedge \sim q) \wedge r$

b $(\sim p \wedge q) \wedge r$

c $(p \wedge q) \vee r$

d $p \implies (q \vee \sim r)$

7

a $(r \wedge q) \implies p$

b $(\sim p \iff \sim q) \vee \sim r$

c $(r \iff q) \wedge p$

d $q \implies (p \iff r)$

8

a The water is 82°F or the sun is down, and we do not go swimming.

- b** The water is not 28°C , and the sun is down  we go swimming.
- c** If we do not go swimming, then the sun is down and the water is 28°C .
- d** If the sun is down then we go swimming, and the water is 28°C .
- e** The sun is down if and only if the water is 28°C , and we go swimming.

9

- a** The water is 60°F and the sun is up, or we go snorkelling.
- b** If the sun is up then the water is 60°F , or we go snorkelling.
- c** If the sun is up and we go snorkelling, then the water is 60°F .
- d** If the water is not 60°F , then the sun is up or we go snorkelling.
- e** If the sun is up, then the water is 60°F if and only if we go snorkelling.

10

- a** $p \wedge \sim q$ **b** Conjunction

11

- a** $p \implies (q \vee r)$ **b** Conditional

12

- a** $\sim (p \implies \sim q)$ **b** Negation

13

- a** $(p \implies q) \vee \sim r$ **b** Disjunction

14

- a** $(p \vee q) \implies r$ **b** Conditional

15

- a** $(p \implies q) \iff \sim r$ **b** Biconditional

16

- a** $(p \wedge q) \implies r$ **b** Conditional

17

- a** $(p \iff q) \vee r$ **b** Disjunction

Additional questions



- 18** A biconditional statement is a statement that combines a conditional statement with its converse. This type of statement is constructed using the phrase "if and only if" which can be used to recognise it.
- 19** The following statements are biconditional
- a** A triangle is equilateral if and only if it is equiangular.
 - c** It is Cinco de Mayo if and only if it is the 5th of May.
- 20** If a ray is an angle bisector then it divides an angle into two congruent angles.
If a ray divides an angle into two congruent angles then it is an angle bisector.
- 21** Two sides of a triangle are congruent if and only if the two angles opposite them are congruent.
- 22** A biconditional statement is true when the two implied conditional statements are true.
- 23**
- a** True. We know this because the two implied conditional statements:
 - If a number is even, then it is divisible by two.
 - If a number is divisible by two, then it is even.are true.
 - b** False. This is because the implied conditional statement:
 - A number is greater than four if it is greater than three.is false, with a counterexample being the number $3\frac{1}{2}$.
 - c** True. We know this because the two implied conditional statements:
 - If a number is prime, then it has no factors other than itself and one.
 - If a number has no factors other than itself and one, then it is prime.are true.
 - d** False. This is because the implied conditional statement:
 - If all the sides of a shape are congruent, then it is a regular polygon.is false, with a counterexample being a rhombus.

If a quadrilateral has diagonals that bisect each other, then it is a parallelogram.



25

- a A segment is a triangle median if and only if its endpoints are a vertex and the midpoint of the opposite side of a triangle.
- b An angle is a right angle if and only if it measures exactly 90° .
- c A three-dimensional figure is a sphere if and only if it consists of all points equidistant from a point called the center.

26 For a biconditional statement to be false, at least one of the implied conditional statements must be false. Consider that:

- a Neva can use an umbrella even if it is not raining.
- b It is possible to see both the sun and the moon at the same time. For example, before and after a solar eclipse.
- c Deloris can choose to walk to school without having to miss the bus.
- d Ulrich could get items from the top shelf without being tall by using a stool or a grabbing tool to reach them.

27

- a Not a valid definition, since it is not reversible. The statement "if the measure of an angle is greater than 90° , then it's obtuse" is not true, so we cannot write this definition as an "if and only if" statement.
- b Valid
- c Not a valid definition, since the phrase "next to each other" is not precise.
- d Not a valid definition, since the terms "vector" and "dot product" are not commonly understood or defined in this course.

28 No. Knowing that a biconditional statement is not true doesn't tell us which of the two implied conditional statements is not true.

29 A valid definition can be written as a biconditional statement because it matches a specific set of properties to an object (or scenario) in an exclusive relationship which can be represented by the phrase "if and only if".

30 0.1